



IILM UNIVERSITY

School of Computer Science & Engineering

SUMMER INTERNSHIP PRESENTATION

CodeAlpha VIP in Data Science

INTERNSHIP TOPIC / TITLE

**Data Science & Machine Learning
Projects: Iris Classification and Car Price Prediction**

• STUDENT	Sweta Prakash
• ENROLLMENT NO.	25SCS1003004266
• PROGRAM & SEMESTER	B.Tech CSE • Semester 3
• ORGANIZATION	Code Alpha Software Development

B.Tech Computer Science & Engineering • IILM University, Greater Noida

INTERNSHIP FOCUS

DS

Data Science

ML

Machine Learning

PY

Python & Libraries

ML

Predictive Modeling

20 July – 20 August 2026
4-week virtual internship

Internship Overview

Program

CodeAlpha VIP Internship Program
Domain: Data Science & Machine Learning

Duration

4 weeks
20 July 2026 – 20 August 2026

Academic Context

B.Tech CSE internship requirement
at IILM University

- The internship focused on applying theoretical Machine Learning concepts to structured tabular datasets.
- Work covered preprocessing, feature engineering, model training, evaluation and visualization.
- Two practical tasks formed the core implementation: Iris Flower Classification and Car Price Prediction.

Organization Profile — CodeAlpha

- Technical organization focused on skill development and hands-on industrial virtual internships.
- Provides structured, project-oriented learning environments for engineering students.
- Learning exposure includes Data Science, Artificial Intelligence and Full-Stack Software Engineering.
- Mission stated in the report: “Your Dream Our Passion”.

Leadership

Ms. Swati Srivastava
Founder & CEO

Learning Approach

Structured projects + practical
problem statements

Location in Report

Lucknow, Uttar Pradesh

Project Objectives

EDA & Cleaning

Use Pandas and NumPy for exploratory analysis and data cleansing.

Preprocessing

Handle outliers, missing values and normalize numerical features.

Classification

Apply Logistic Regression, SVC, Random Forest and k-NN.

Regression

Build Linear, Ridge and Gradient Boosting regression models.

Evaluation

Use Accuracy, F1, Confusion Matrix, MAE, RMSE and R^2 .

Technologies & Tools

Programming

Python 3.10+

Data

Pandas, NumPy

ML

Scikit-learn

Visualization

Matplotlib, Seaborn

Development

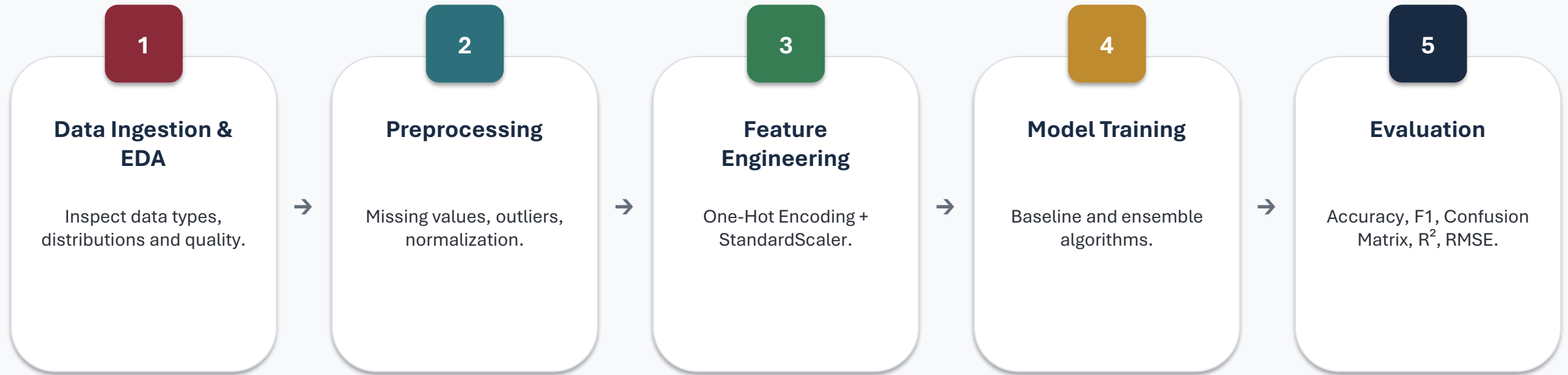
Jupyter Notebook, Visual Studio
Code

Version Control

Git, GitHub

Open-source Python ecosystem used from data preparation through model evaluation and visualization.

System Architecture / End-to-End Workflow



RAW DATA → PREPROCESSING → FEATURES → MODELS → INSIGHTS

Task 1 — Iris Flower Classification

Dataset

Features:

- Sepal Length (cm)
- Sepal Width (cm)
- Petal Length (cm)
- Petal Width (cm)

Target:

Setosa, Versicolor, Virginica

Methodology

- Exploratory Data Analysis with scatter plots and pairplots.
- StandardScaler for feature normalization.
- Multiple supervised classifiers trained and compared.

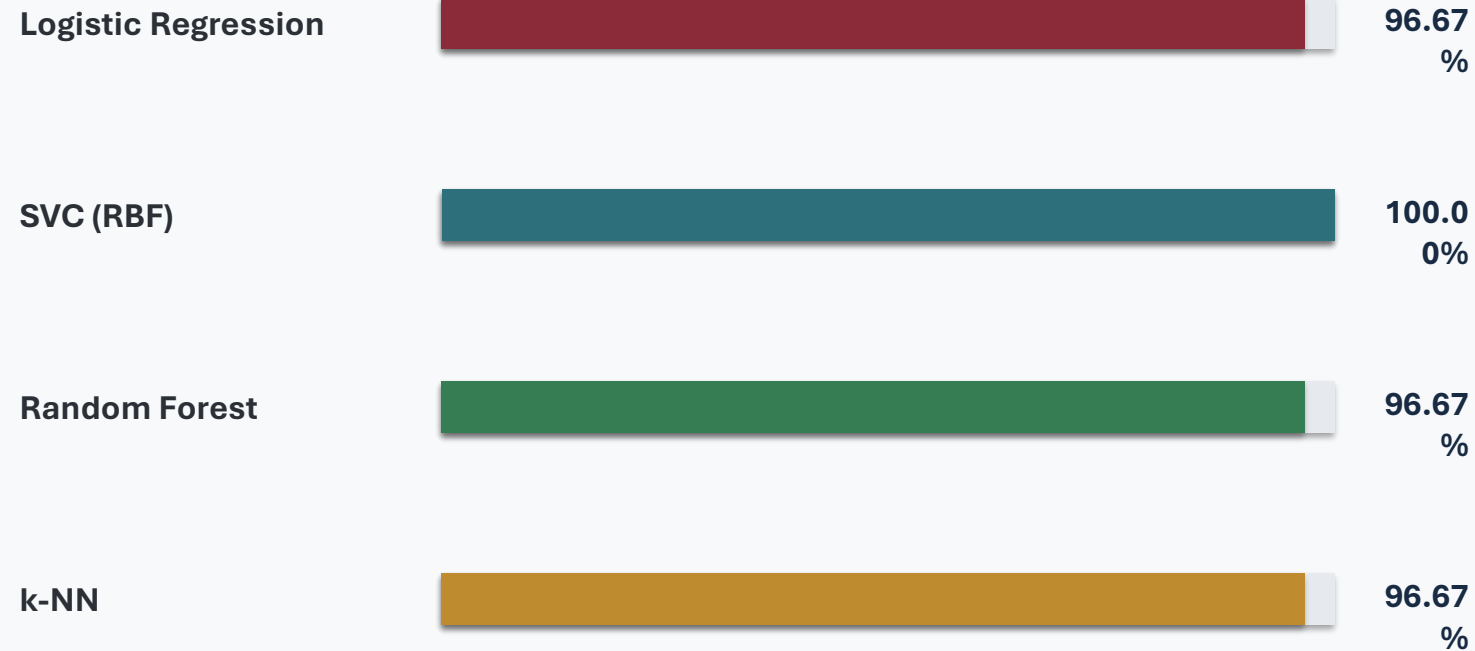
Important Observation

Setosa was reported as linearly separable, while Versicolor and Virginica required more complex decision boundaries.

Best reported model: SVC with RBF kernel.

Task 1 — Classification Results

Testing Accuracy



Best Model

Support Vector Classifier
(RBF)

Testing Accuracy: 100.00%
Macro F1: 1.000

Model Comparison

Logistic Regression: 96.67%
Random Forest: 96.67%
k-NN: 96.67% testing
accuracy

Task 2 — Car Price Prediction

Brand / Make

Categorical market and reputation factor.

Horsepower & Engine Size

Numerical indicators linked to performance and price class.

Mileage & Fuel Economy

Fuel-efficiency variables with reported inverse pricing relationship.

Vehicle Age / Model Year

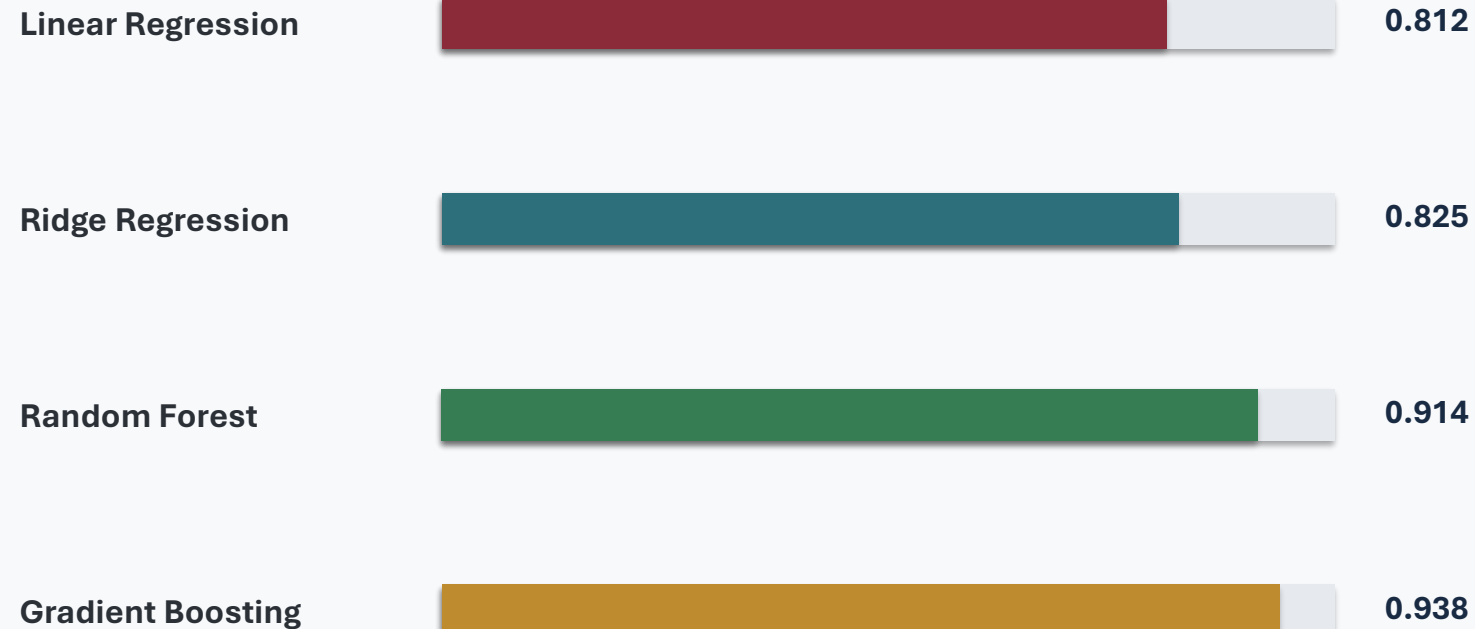
Captures depreciation across vehicle lifespan.

Methodology

One-Hot Encoding for categorical variables • log-transform of target price to reduce skewness • compare Linear Regression, Ridge, Random Forest and Gradient Boosting.

Task 2 — Regression Results

R² Score (higher is better)



Best Performer

Gradient Boosting
Regressor

R² = 0.938

RMSE = \$1,420.15

MAE = \$980.20

Interpretation

Gradient Boosting achieved the strongest reported regression performance and the lowest reported error values.

Overall Results at a Glance

Iris Classification

Best model: SVC (RBF)

Testing Accuracy: 100.00%

Macro F1-Score: 1.000

The classifier successfully separated the three species on the reported test split.

Car Price Prediction

Best model: Gradient Boosting Regressor

R^2 Score: 0.938

RMSE: \$1,420.15

MAE: \$980.20

The ensemble model gave the strongest reported fit among evaluated regressors.

Learning Outcomes

Data Preparation

Clean, preprocess, scale and transform structured data.

Feature Engineering

Encode categorical features and create model-ready inputs.

Model Experimentation

Compare baseline, regularized and ensemble learning methods.

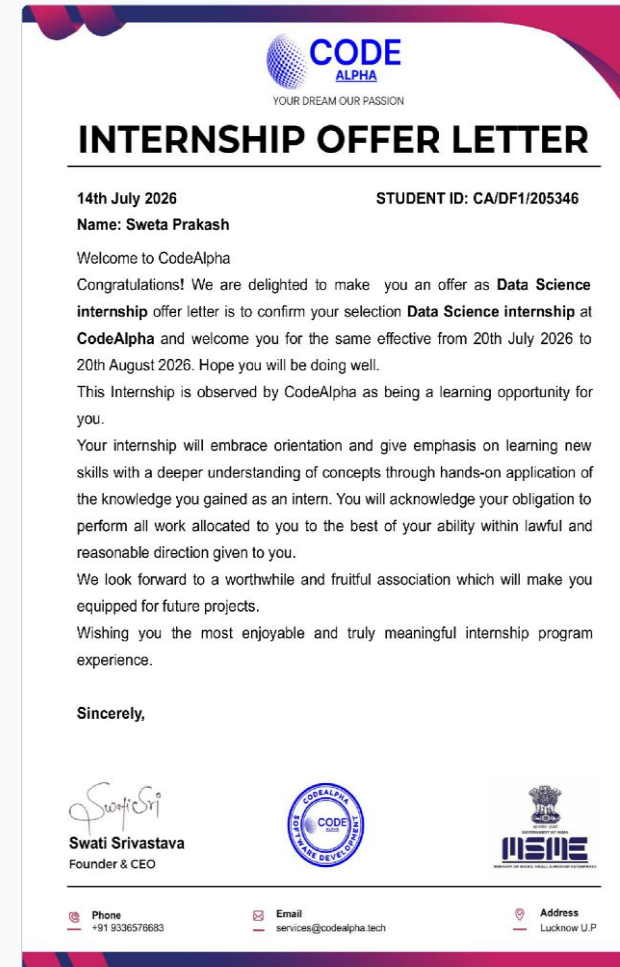
Model Evaluation

Interpret Accuracy, F1, confusion matrix, MAE, RMSE and R^2 .

Internship Documentation



Certificate of Completion — issued 20 August 2026



Internship Offer Letter

Academic Record

Internship period:
20 Jul – 20 Aug
2026

Domain:
Data Science

Student ID:
CA/DF1/205346

Conclusion

- The internship provided practical exposure to the end-to-end Data Science and Machine Learning workflow.
- The two projects demonstrated supervised learning in both classification and regression settings.
- Iris classification achieved the strongest reported test result with SVC (RBF): 100% testing accuracy.
- Car price prediction achieved the strongest reported regression result with Gradient Boosting: $R^2 = 0.938$.
- The work strengthened practical skills in preprocessing, feature engineering, model comparison and statistical evaluation.

Overall Outcome: Applied understanding of Data Science & Machine Learning

References

- CodeAlpha Software Development — Internship Offer Letter & Completion Guidelines (2026).
- CodeAlpha Software Development — Certificate of Completion, Data Science Virtual Internship Program (2026).
- Pedregosa, F. et al. — Scikit-learn: Machine Learning in Python, JMLR, 2011.
- McKinney, W. — Data Structures for Statistical Computing in Python, 2010.
- Fisher, R. A. — The Use of Multiple Measurements in Taxonomic Problems, 1936.
- Hastie, Tibshirani & Friedman — The Elements of Statistical Learning, 2nd ed., Springer, 2009.
- Hunter, J. D. — Matplotlib: A 2D Graphics Environment, 2007.

Thank You

Questions & Discussion

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